S3 Table.

Results for every arm of every experiment

Supporting information for "Environmental fluctuation limits gene regulation unless it can be tracked"

Values are means ± standard deviations over populations, the unit of analysis. They are generated by `paper/supplement/make_s3.py` from the per-run files `paper/data/evosim_runs.csv`, `raevol_runs.csv`, `grain_runs.csv`, `decomp_runs.csv`, `confirm_runs.csv` and `switch_trajectories.csv`, which contain one row per run with the same summary measures used in the pre-registered analyses (the final 20% of each run for EvoSim and generations 25,001–30,000 for R-Aevol, except the exploratory decomposition, which uses generations 30,001–32,000 of continued runs, and the environment switch, whose windows are given with its table). R-Aevol values describe the best individual in each population, as recorded by the platform. Total regulatory strength S is the number of regulatory links times their mean strength, and is zero for generations with no links.

Table 1. EvoSim, 180 runs (grid 128, 150,000 steps, 30 seeds per arm). Populations that went extinct are counted and excluded from the means. The three frozen arms are identical runs (the cue is zero with a single state) and are listed once.

Arm	Survived	Count density	Strength density	Total strength	Nodes	Population
Fluctuating, informative	29/30	0.550 ± 0.058	0.190 ± 0.030	40.1 ± 12.2	14.38 ± 1.20	9865 ± 633
Fluctuating, yoked	29/30	0.597 ± 0.073	0.190 ± 0.026	39.3 ± 11.1	14.24 ± 1.26	9912 ± 677
Fluctuating, no cue	28/30	0.587 ± 0.065	0.184 ± 0.019	37.5 ± 11.9	14.11 ± 1.37	9899 ± 609
Frozen (all three arms)	30/30	0.467 ± 0.040	0.207 ± 0.014	38.0 ± 5.3	13.53 ± 0.78	16568 ± 2825

Time steps per generation, from the slope of mean lineage depth over the second half of each run: fluctuating arms 23.6–24.3, frozen 30.7.

Table 2. R-Aevol, exploratory experiment (72 runs; $q = 0.10$, population 1024, 30,000 generations).

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, informative	12	12.68 ± 3.18	69.6 ± 19.9	19.00 ± 3.18	0.0288 ± 0.0025
Fluctuating, yoked	12	5.66 ± 1.52	39.6 ± 7.8	17.86 ± 1.91	0.0584 ± 0.0021
Fluctuating, no cue	12	1.42 ± 0.76	15.0 ± 6.6	18.85 ± 2.61	0.0626 ± 0.0005
Frozen, informative	12	4.69 ± 2.47	80.0 ± 45.3	26.12 ± 4.45	0.0078 ± 0.0014
Frozen, no cue	12	8.21 ± 5.09	68.8 ± 51.9	26.75 ± 3.96	0.0082 ± 0.0012
Frozen, yoked	12	6.17 ± 7.12	76.8 ± 63.0	24.64 ± 2.00	0.0087 ± 0.0010

Table 3. R-Aevol, pre-registered sweep (216 runs, 30,000 generations; population 1024 unless marked).

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, informative, $q = 0.05$	12	11.03 ± 4.38	56.4 ± 22.9	16.34 ± 5.69	0.0382 ± 0.0361
Fluctuating, yoked, $q = 0.05$	12	6.32 ± 2.46	39.9 ± 10.6	16.36 ± 2.15	0.0547 ± 0.0032
Fluctuating, no cue, $q = 0.05$	12	1.11 ± 0.83	11.4 ± 7.2	17.45 ± 1.97	0.0601 ± 0.0012
Frozen, informative, $q = 0.05$	12	5.95 ± 3.21	94.0 ± 46.6	26.28 ± 3.48	0.0077 ± 0.0013
Fluctuating, informative, $q = 0.10$	12	13.39 ± 2.34	71.2 ± 18.6	18.07 ± 2.95	0.0293 ± 0.0035
Fluctuating, yoked, $q = 0.10$	12	5.73 ± 1.71	41.6 ± 9.5	19.94 ± 3.83	0.0583 ± 0.0021
Fluctuating, no cue, $q = 0.10$	12	1.25 ± 0.71	12.7 ± 5.1	20.03 ± 3.44	0.0627 ± 0.0007
Frozen, informative, $q = 0.10$	12	7.53 ± 4.89	91.0 ± 39.9	28.28 ± 2.69	0.0081 ± 0.0017
Fluctuating, informative, $q = 0.20$	12	13.18 ± 2.05	69.3 ± 18.0	18.02 ± 2.50	0.0303 ± 0.0030
Fluctuating, yoked, $q = 0.20$	12	4.56 ± 2.38	38.3 ± 15.0	20.75 ± 2.14	0.0626 ± 0.0016
Fluctuating, no cue, $q = 0.20$	12	1.14 ± 0.84	12.4 ± 6.4	22.43 ± 3.12	0.0641 ± 0.0004
Frozen, informative, $q = 0.20$	12	4.80 ± 2.50	83.2 ± 55.4	25.60 ± 4.03	0.0085 ± 0.0016
Frozen, no cue, $q = 0.10$	12	6.30 ± 3.09	62.4 ± 26.3	27.02 ± 3.97	0.0083 ± 0.0013
Frozen, yoked, $q = 0.10$	12	5.39 ± 1.93	86.4 ± 50.1	28.02 ± 7.27	0.0079 ± 0.0012
Fluctuating, informative, $q = 0.10$, $N = 512$	12	10.55 ± 2.50	53.3 ± 18.9	17.47 ± 3.13	0.0349 ± 0.0054
Fluctuating, yoked, $q = 0.10$, $N = 512$	12	4.29 ± 2.78	28.9 ± 12.9	17.92 ± 2.26	0.0609 ± 0.0022
Fluctuating, no cue, $q = 0.10$, $N = 512$	12	1.63 ± 0.93	16.5 ± 5.9	20.22 ± 3.53	0.0635 ± 0.0006
Frozen, informative, $q = 0.10$, $N = 512$	12	4.81 ± 2.40	59.8 ± 18.5	25.29 ± 3.08	0.0098 ± 0.0022

Table 4. R-Aevol, grain experiment (60 runs, no cue, population 1024, 30,000 generations). Mean dwell is $1/q$ ages; 20 ages make one generation.

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, $q = 0.1$ (dwell 10 ages)	12	2.29 ± 1.32	17.8 ± 7.5	19.45 ± 2.22	0.0624 ± 0.0010
Fluctuating, $q = 0.01$ (dwell 100 ages)	12	1.06 ± 0.57	9.5 ± 3.1	16.25 ± 1.85	0.0594 ± 0.0018
Fluctuating, $q = 0.001$ (dwell 1,000 ages)	12	1.17 ± 0.60	13.1 ± 6.4	17.36 ± 0.95	0.0517 ± 0.0017
Fluctuating, $q = 0.0002$ (dwell 5,000 ages)	12	2.04 ± 0.98	22.9 ± 9.5	19.24 ± 1.83	0.0376 ± 0.0014
Frozen	12	8.40 ± 3.30	67.8 ± 26.6	25.58 ± 4.50	0.0081 ± 0.0015

Table 5. R-Aevol, signal-link decomposition (exploratory; 252 runs continued from their saved generation-30,000 populations to 32,000, population 1024 unless stated). S_{sig} is the part of the regulatory strength exerted by signal proteins, S_{int} the genome-encoded part.

Experiment	Cell	n	S	S_{sig}	S_{int}	Signal links	G
Sweep	Fluctuating, informative, $q = 0.05$	12	11.28 ± 4.75	5.84 ± 1.98	5.45 ± 3.49	18.1 ± 6.4	4
Sweep	Fluctuating, yoked, $q = 0.05$	12	6.64 ± 2.36	4.50 ± 1.66	2.15 ± 1.64	25.4 ± 7.1	
Sweep	Fluctuating, no cue, $q = 0.05$	12	1.28 ± 0.70	0.00 ± 0.00	1.28 ± 0.70	0.0 ± 0.0	
Sweep	Frozen, informative, $q = 0.05$	12	5.97 ± 2.85	1.13 ± 2.07	4.84 ± 1.99	52.1 ± 37.5	4
Sweep	Fluctuating, informative, $q = 0.10$	12	13.06 ± 2.16	7.20 ± 1.77	5.87 ± 1.77	22.3 ± 5.4	4
Sweep	Fluctuating, yoked, $q = 0.10$	12	6.29 ± 2.19	4.71 ± 2.22	1.59 ± 0.65	27.8 ± 8.5	
Sweep	Fluctuating, no cue, $q = 0.10$	12	1.08 ± 0.63	0.00 ± 0.00	1.08 ± 0.63	0.0 ± 0.0	
Sweep	Frozen, informative, $q = 0.10$	12	7.17 ± 4.45	1.75 ± 2.90	5.42 ± 2.50	48.6 ± 49.0	4
Sweep	Fluctuating, informative, $q = 0.20$	12	13.45 ± 2.01	6.66 ± 1.17	6.79 ± 1.63	20.1 ± 3.8	5
Sweep	Fluctuating, yoked, $q = 0.20$	12	4.76 ± 2.33	3.81 ± 2.36	0.95 ± 0.58	25.7 ± 12.0	
Sweep	Fluctuating, no cue, $q = 0.20$	12	1.05 ± 0.61	0.00 ± 0.00	1.05 ± 0.61	0.0 ± 0.0	
Sweep	Frozen, informative, $q = 0.20$	12	5.09 ± 2.59	0.65 ± 1.39	4.44 ± 2.16	46.5 ± 52.7	3
Sweep	Frozen, yoked, $q = 0.10$	12	5.31 ± 1.61	0.42 ± 0.75	4.89 ± 1.51	58.7 ± 57.3	4
Sweep	Frozen, no cue, $q = 0.10$	12	6.44 ± 3.24	0.00 ± 0.00	6.44 ± 3.24	0.0 ± 0.0	6
Sweep	Fluctuating, informative, $q = 0.10$, population 512	12	10.47 ± 2.55	5.99 ± 0.96	4.48 ± 2.55	18.9 ± 2.8	3
Sweep	Fluctuating, yoked, $q = 0.10$, population 512	12	4.17 ± 2.73	3.03 ± 2.43	1.14 ± 0.67	19.0 ± 9.7	
Sweep	Frozen, informative, $q = 0.10$, population 512	12	4.75 ± 2.86	1.09 ± 2.37	3.67 ± 1.37	18.4 ± 10.0	3
Exploratory	Fluctuating, informative, $q = 0.10$	12	13.11 ± 3.74	6.76 ± 1.58	6.35 ± 3.40	20.9 ± 4.3	5
Exploratory	Frozen, informative, $q = 0.10$	12	4.72 ± 2.66	0.06 ± 0.06	4.66 ± 2.66	29.6 ± 38.5	3
Exploratory	Fluctuating, yoked, $q = 0.10$	12	5.83 ± 1.91	4.16 ± 2.03	1.67 ± 0.87	24.3 ± 7.2	
Exploratory	Frozen, yoked, $q = 0.10$	12	4.86 ± 3.79	1.11 ± 2.66	3.75 ± 1.87	40.9 ± 49.3	3

Table 6. R-Aevol, pre-registered confirmation (132 runs with the instrumented platform from generation 0, population 1024, 30,000 generations; means over generations 25,001–30,000).

Cell	n	S	S_{sig}	S_{int}	Links	Genes	Metabolic error
Fluctuating, informative, $q = 0.10$	12	13.85 ± 2.26	7.32 ± 1.51	6.53 ± 2.47	73.6 ± 23.8	18.73 ± 4.12	0.0294 ± 0.0036
Fluctuating, yoked, $q = 0.10$	12	5.60 ± 2.47	3.76 ± 1.48	1.84 ± 1.87	38.5 ± 14.1	18.66 ± 3.25	0.0586 ± 0.0024
Fluctuating, no cue, $q = 0.10$	12	1.34 ± 0.84	0.00 ± 0.00	1.34 ± 0.84	12.4 ± 6.4	20.26 ± 2.63	0.0624 ± 0.0005
Frozen, informative, $q = 0.10$	12	4.81 ± 2.94	0.72 ± 2.11	4.09 ± 2.03	64.6 ± 36.1	26.34 ± 3.65	0.0084 ± 0.0013
Frozen, yoked, $q = 0.10$	12	4.56 ± 2.82	0.53 ± 1.43	4.03 ± 1.90	81.9 ± 66.1	28.24 ± 4.70	0.0074 ± 0.0009
Fluctuating, informative, $q = 0.20$	12	12.82 ± 2.98	6.85 ± 0.83	5.96 ± 2.68	71.4 ± 23.6	18.95 ± 2.80	0.0315 ± 0.0029
Fluctuating, yoked, $q = 0.20$	12	5.19 ± 2.33	3.99 ± 1.67	1.20 ± 1.18	35.7 ± 12.3	20.55 ± 2.10	0.0625 ± 0.0007
Fluctuating, no cue, $q = 0.20$	12	1.31 ± 1.01	0.00 ± 0.00	1.31 ± 1.01	13.3 ± 8.8	22.60 ± 4.54	0.0643 ± 0.0005
Frozen, informative, $q = 0.20$	12	4.79 ± 2.28	0.91 ± 1.93	3.88 ± 1.58	73.4 ± 42.2	27.49 ± 4.86	0.0084 ± 0.0012
Frozen, yoked, $q = 0.20$	12	6.17 ± 2.09	0.53 ± 0.56	5.64 ± 1.79	74.0 ± 33.8	26.93 ± 4.56	0.0086 ± 0.0012
Frozen, no cue	12	7.88 ± 2.41	0.00 ± 0.00	7.88 ± 2.41	68.7 ± 25.7	27.04 ± 3.70	0.0079 ± 0.0010

Table 7. R-Aevol, pre-registered environment switch (48 runs, 12 per cell, each founded by the best genome of a cue-free confirmation population at $q = 0.10$, generation 30,000, and run for 10,000 generations; source over generations 25,001–30,000, start over 1–500, end over 8,001–10,000; no signals, so S is the genome-encoded network).

Cell	n	S source	S start	S end	Regulators end	Genes end	Metabolic error end
Frozen, kept frozen	12	7.88 ± 2.41	7.96 ± 2.49	8.49 ± 2.59	17.7 ± 3.7	27.3 ± 3.8	0.0075 ± 0.0011
Frozen, moved to fluctuation	12	7.88 ± 2.41	4.71 ± 1.44	1.79 ± 0.53	8.6 ± 2.2	20.2 ± 2.4	0.0623 ± 0.0007
Fluctuating, kept fluctuating	12	1.34 ± 0.84	1.51 ± 1.02	1.28 ± 0.80	6.7 ± 2.3	19.9 ± 3.0	0.0622 ± 0.0009
Fluctuating, moved to frozen	12	1.34 ± 0.84	2.52 ± 1.74	5.25 ± 2.13	15.6 ± 2.8	27.3 ± 3.2	0.0093 ± 0.0012